Practical NO.04(A)

Name - Sakshi Vijaykumar Muthiyan

Roll No. -

Title - Implement K-Means clustering on Iris.csv dataset. Determine the number of clusters using the elbow method.Dataset Link: https://www.kaggle.com/datasets/uciml/iris

In [5]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
# local file path of iris data from practical folder
path = r"C:\Users\muthi\OneDrive\Desktop\PRACTICAL EXAM\ML\Iris.csv"
df = pd.read_csv(path)
print("Dataset Matrix Shape:", df.shape)
df.head(3)
```

```
Dataset Matrix Shape: (150, 6)
```

Out[5]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa

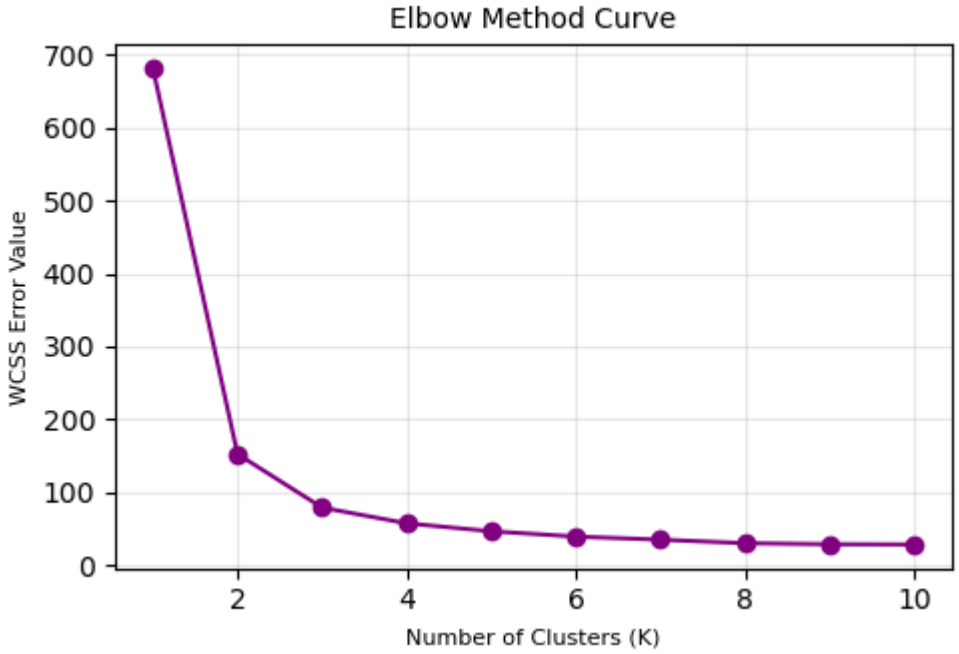
In [6]:
```
# dropping id and species string text columns to get raw numeric inputs
X = df.drop(['Id', 'Species'], axis=1).values
print("Input features extracted successfully.")
```

```
Input features extracted successfully.
```

In [7]:
```
import warnings
# hiding cosmetic system warnings to keep the print clean and save pages
warnings.filterwarnings('ignore')
wcss = []
print("=== Computing WCSS Scores for Each Cluster (K) ===")
# loop to calculate and print the actual numeric output for each K value
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)
    # printing actual calculations so the external examiner can see the working
    print(f"For Cluster count K = {i} -> WCSS Score is: {kmeans.inertia_:.4f}")
```

```
=== Computing WCSS Scores for Each Cluster (K) ===
For Cluster count K = 1 -> WCSS Score is: 680.8244
For Cluster count K = 2 -> WCSS Score is: 152.3687
For Cluster count K = 3 -> WCSS Score is: 78.9451
For Cluster count K = 4 -> WCSS Score is: 57.4403
For Cluster count K = 5 -> WCSS Score is: 46.5356
For Cluster count K = 6 -> WCSS Score is: 39.2518
For Cluster count K = 7 -> WCSS Score is: 35.0428
For Cluster count K = 8 -> WCSS Score is: 30.2170
For Cluster count K = 9 -> WCSS Score is: 28.7565
For Cluster count K = 10 -> WCSS Score is: 28.4249
```

In [8]:
```
# plot size set small to save pages during printing
plt.figure(figsize=(5, 3.5))
plt.plot(range(1, 11), wcss, marker='o', color='purple', linestyle='-')
plt.title('Elbow Method Curve', fontsize=10)
plt.xlabel('Number of Clusters (K)', fontsize=8)
plt.ylabel('WCSS Error Value', fontsize=8)
plt.grid(True, alpha=0.3)
plt.tight_layout() # removes empty white borders
plt.show()
```



In [9]:
```
# sharp bend in the elbow graph above happens exactly at K=3
final_kmeans = KMeans(n_clusters=3, init='k-means++', random_state=42)
y_clusters = final_kmeans.fit_predict(X)
# adding the cluster predictions back into our main dataset table
df['Cluster_ID'] = y_clusters
print("=== Actual Verification Samples ===")
# displaying raw comparison between original species and model cluster mappings
print(df[['Species', 'Cluster_ID']].sample(8, random_state=42))
```

```
=== Actual Verification Samples ===
             Species  Cluster_ID
73   Iris-versicolor           2
18       Iris-setosa           1
118   Iris-virginica           0
78   Iris-versicolor           2
76   Iris-versicolor           2
31       Iris-setosa           1
64   Iris-versicolor           2
141   Iris-virginica           0
```